

Discrete Structures

Homework Set 1 • Propositional & Predicate Logic

Due: Friday, Feb 7, 2026 • 11:00 PM (ET)

Homework 1

Submit as a single PDF export of your typed work

Format & Integrity. Write complete solutions in your own words. If you consulted any external resources (notes, websites, AI tools, tutors), disclose them clearly for each problem. Do not post these problems or your solutions publicly.

SUBMISSION GUIDELINES

- Type your solutions. Keep your work legible and well-organized.
- Start each problem on a new page in your final PDF.
- When using equivalence transformations, state the law/rule you are applying.
- Unless explicitly allowed, do not use truth tables when asked to use algebraic equivalences.

Problem 1 — Required Statement

Required

Type the following sentence and replace the bracketed text with your full name.

I, [YOUR FULL NAME], affirm that I completed this assignment without unauthorized help. I wrote my solutions in my own words and I have clearly cited any permitted resources I used.

Problem 2 — Tautology, Contradiction, or Neither

3 × 3 pts

For each formula, determine whether it is a tautology, a contradiction, or neither. **Do not** use truth tables. If it is a tautology/contradiction, prove it using equivalence rules (name the rules). If neither, give a counterexample assignment.

a. $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

b. $\neg(p \wedge (q \vee r)) \wedge (p \rightarrow q)$

c. $(p \leftrightarrow q) \wedge (q \leftrightarrow \neg p)$

Hint: Common tools include implication elimination ($p \rightarrow q \equiv \neg p \vee q$), De Morgan's laws, distributivity, absorption, and contradiction/tautology identities.

Problem 3 — Satisfiable or Unsatisfiable

3 × 3 pts

For each statement: either (i) show it is unsatisfiable via equivalences (name the equivalences), or (ii) give an explicit satisfying assignment.

a. $\neg(p \vee s) \wedge (p \vee s)$

b. $(a \vee b) \wedge (\neg a \vee c) \wedge (\neg b \vee c) \wedge \neg c$

c. $(x \rightarrow y) \wedge (y \rightarrow x) \wedge (x \vee y)$

Problem 4 — Predicate Logic Equivalences

3 × 3 pts

State whether each pair of statements is logically equivalent. If yes, justify (briefly but clearly). If not, give a counterexample structure/interpretation.

- a. $\forall x (P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$
- b. $\exists x (P(x) \vee Q(x))$ and $\exists x P(x) \vee \exists x Q(x)$
- c. $\forall x \exists y R(x, y)$ and $\exists y \forall x R(x, y)$

Tip: Watch quantifier order carefully. Swapping $\forall$ and $\exists$ is usually not valid.

Problem 5 — Formalizing an English Statement

2 × 4 pts

Universe: all integers. Let $N(x)$ mean “ x is negative” and $G(x)$ mean “ x is greater than 5.” Consider the English statement:

“No negative integer is greater than 5.”

For each formula below, decide whether it correctly expresses the statement, and justify.

- a. $\neg \exists x (N(x) \wedge G(x))$
- b. $\forall x (N(x) \rightarrow \neg G(x))$

Your justification can be short: a 1–3 sentence logical explanation or a concrete counterexample.

Problem 6 — Push Negations Inward

14 pts

Rewrite the following formula so that every negation symbol ($\neg$) appears immediately in front of a predicate (i.e., no negation should apply to a compound formula or quantifier). Show your steps.

$\neg \exists x \forall y \exists z (A(x, y) \vee \neg B(y, z) \wedge C(z))$

Note: Use quantifier-negation rules (e.g., $\neg \forall x \phi \equiv \exists x \neg \phi$), De Morgan's laws, and remember $\wedge$ binds tighter than $\vee$.